Maharashtra Institute of Technology

Vishwaprayag University, Solapur

Department of Computer Science and Engineering

Weekly Progress Report

Weeks 1 – 3

PulseTech — Pediatric Pharmacovigilance Dashboard
using Heterogeneous Graph Attention Networks (HANConv)
for Proactive Adverse Drug Reaction Detection

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Table of Contents

1. Project Overview & Technology Stack
2. Week 1 — Data Pipeline & Database Architecture
3. Week 2 — Signal Detection Engine & GNN Implementation
4. Week 3 — Dashboard Integration & PLOS ONE Submission
5. Current System Status
6. Key Metrics & Achievements
7. Dashboard Screenshots
8. Next Steps

1. Project Overview

Project Title: PulseTech — Pediatric Pharmacovigilance Dashboard using Heterogeneous Graph Attention Networks (HANConv) for Proactive Adverse Drug Reaction Detection

Objective: Design and develop a web-based pediatric pharmacovigilance dashboard that combines retrospective Disproportionality Analysis (PRR, ROR, IC, EBGM) with proactive Graph Neural Network (HANConv) signal prediction on the FDA FAERS database, with full data traceability through a 13-step human-gated ingestion pipeline.

Technology Stack

Component	Technology
Frontend	React.js with interactive D3.js / Force-directed graph visualisations
Backend	Python FastAPI with SQLAlchemy ORM
Database	PostgreSQL (faers_pediatric analytical store)
GNN Framework	PyTorch Geometric — HANConv (Heterogeneous Attention Network)
Drug Normalisation	RxNorm API (NIH NLM) for standardised drug naming
ADR Coding	MedDRA v28.1 Preferred Terms (via FAERS REAC files)
Report Generation	In-dashboard one-click PDF export

2. Week 1 — Data Pipeline & Database Architecture

Period: 15 September – 21 September 2026

Tasks Completed

#	Task	Status	Details
1	13-Step FAERS Ingestion Pipeline	✓ Complete	Automated pipeline: file detection → deduplication → demographic normalisation → ICH E11 age banding → RxNorm drug standardisation → human quality gate
2	PostgreSQL Database Setup	✓ Complete	faers_pediatric DB with 7 normalised tables (demo, drug, reac, outc, rpsr, ther, indi)
3	20-Quarter Data Load	✓ Complete	Q1/2021 – Q4/2025 = 3,420,201 patient records
4	ICH E11(R1) Age Banding	✓ Complete	5 bands: Neonate (<28 d), Infant (<2 y), Child (<12 y), Adolescent (<18 y), Unknown
5	RxNorm Drug Normalisation	✓ Complete	~94% drug names resolved via NLM RxNorm API; persistent cache
6	Human Quality Gate (Step 13)	✓ Complete	Mandatory preview / approve before any DB write
7	Upload Hub UI Module	✓ Complete	Drag-and-drop 7-file upload with live pipeline progress and 20-quarter load ledger

Key Metrics

- **Records:** 3,420,201 patient records across 20 quarters
- **Drugs:** 1,902 unique suspect drugs (RxNorm-normalised)
- **ADRs:** 4,416 unique preferred terms (MedDRA v28.1)
- **Age Distribution:** Neonate 23,704 | Infant 29,963 | Child 149,389 | Adolescent 403,278 | Unknown 2,813,867

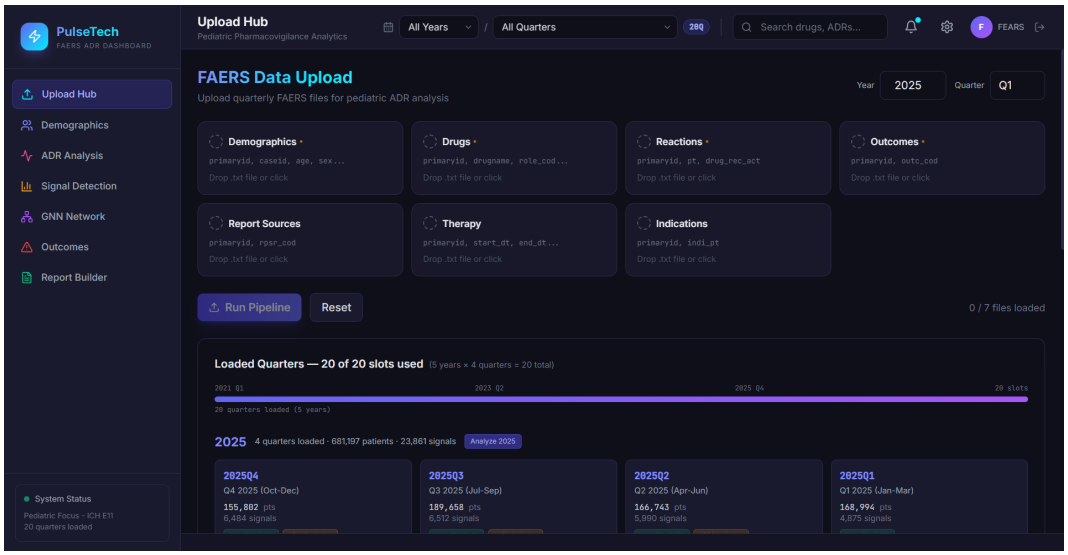


Fig 1 — Upload Hub: 13-step data ingestion pipeline with 20-quarter load ledger

3. Week 2 — Signal Detection Engine & GNN Implementation

Period: 22 September – 28 September 2026

Tasks Completed

#	Task	Status	Details
1	Disproportionality Analysis Engine	✓ Complete	Four gold-standard metrics: PRR, ROR, IC (BCPNN), EBGM
2	Signal Detection Module UI	✓ Complete	In-dashboard signal table with automatic Signal / No-Signal flagging
3	HANConv GNN Architecture	✓ Complete	PediatricADRHAN: 2-layer HANConv, 64-dim hidden, 4-head attention, Adam, 200 epochs
4	Heterogeneous Graph Construction	✓ Complete	CHILD-band: 79 drug nodes, 259 ADR nodes, 500 edges — 419 (83.8%) confirmed signals
5	GNN Network View Module	✓ Complete	Interactive force-directed graph; Classical / GNN Predicted / Both toggle; hover tooltips
6	GNN Link Prediction	✓ Complete	Novel predictions: Dupixent → Acute Pulmonary Oedema (0.87), idursulfase → Abdominal Neoplasm (0.82)
7	Demographics Module	✓ Complete	Age–sex distribution pyramid, sex donut chart, ICH E11 age band breakdown
8	ADR Analysis Module	✓ Complete	Top-20 ADRs stacked bar, ADR × Age heatmap, drug–ADR co-occurrence table

Key Metrics

- **Total Signals:** 82,650 confirmed disproportionality signals across all age bands
- **GNN Graph:** 79 drug nodes, 259 ADR nodes, 500 edges (CHILD band)
- **Signal Density:** 419 / 500 edges (83.8%) confirmed as statistical signals
- **Model:** 2-layer HANConv, 64-dim hidden, 4-head attention, 200 epochs

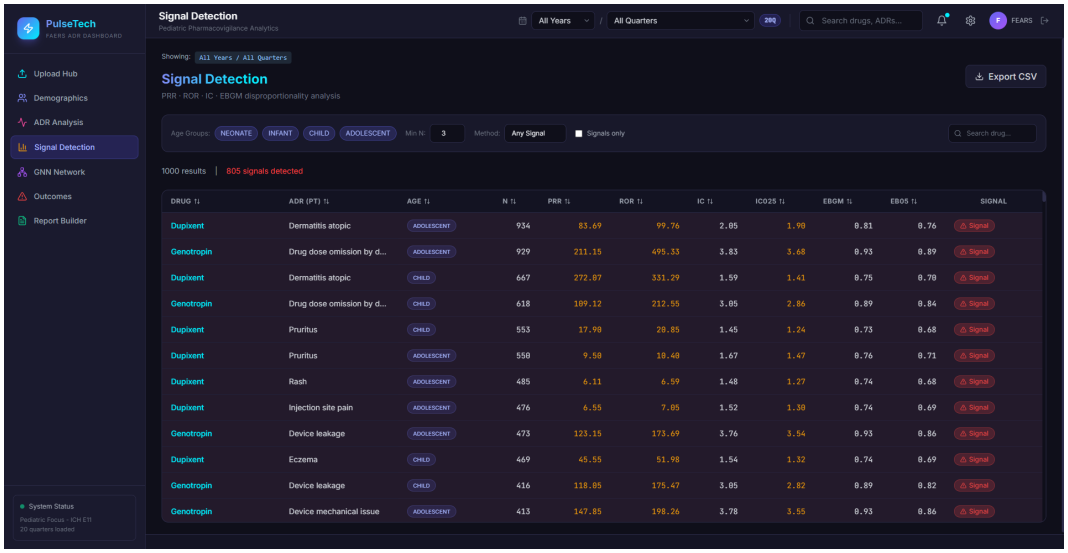


Fig 2 — Signal Detection: PRR / ROR / IC / EBGM disproportionality analysis with signal flagging

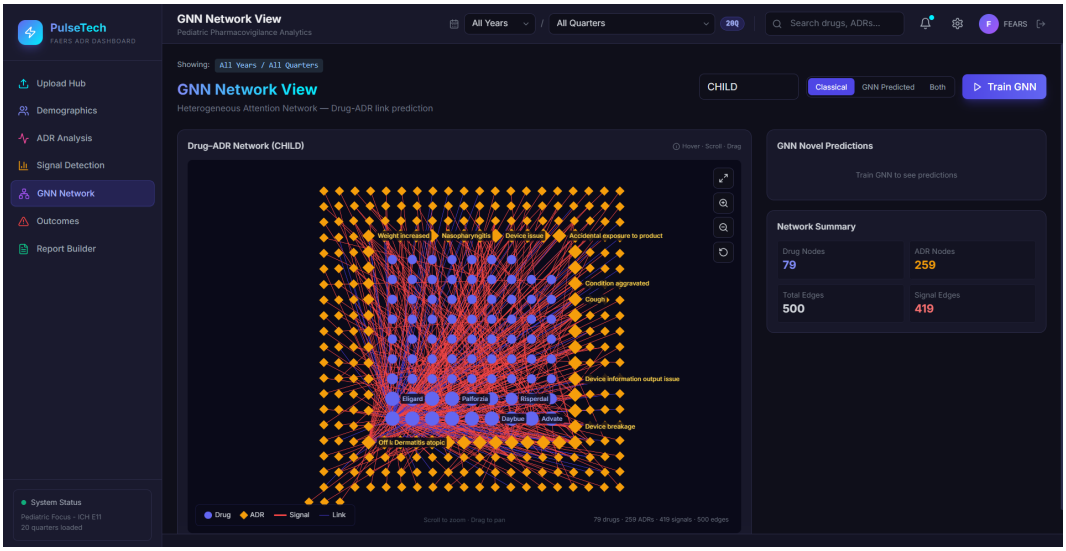


Fig 3 — GNN Network View: Heterogeneous bipartite graph (drug ● and ADR ◆ nodes). Red = confirmed signal edges, Green = GNN predicted

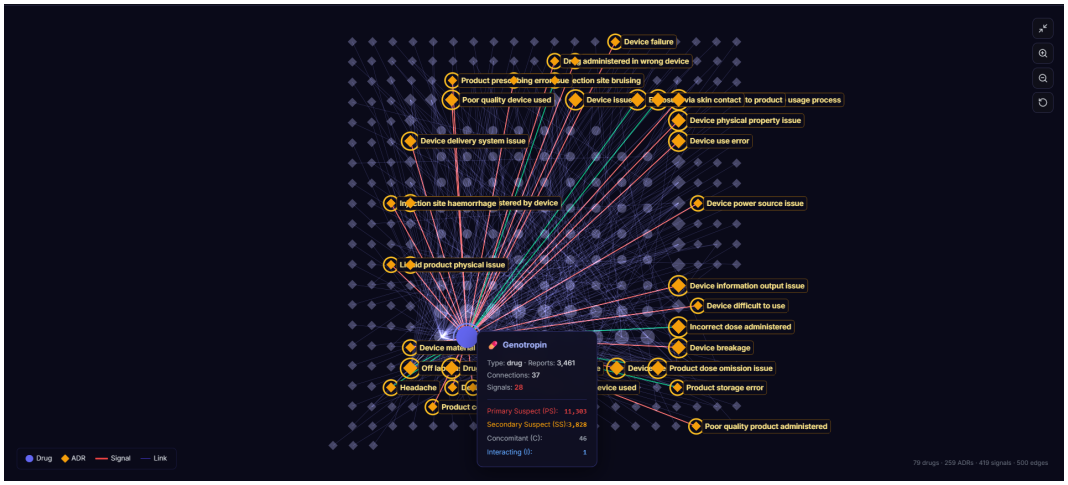


Fig 4 — GNN Node Detail: Genotropin tooltip (3,461 reports, 37 connections, 28 signals; PS: 11,303, SS: 3,828)

4. Week 3 — Dashboard Integration & PLOS ONE Submission

Period: 29 September – 05 October 2026

Tasks Completed

#	Task	Status	Details
1	Outcomes & Severity Module	✓ Complete	Seven-category outcome distribution (Death, Hospitalisation, Disability, Life-Threatening, etc.)
2	Report Builder Module	✓ Complete	Section-selectable one-click PDF export with live preview (<60 s)
3	Full Dashboard Integration	✓ Complete	All 7 modules operational: Upload Hub, Demographics, ADR Analysis, Signal Detection, GNN Network, Outcomes, Report Builder
4	Research Paper (PLOS ONE)	✓ Complete	845-line LaTeX manuscript, 9 figures, 13 tables, 32 references, 244-word abstract
5	Supporting Information S1–S8	✓ Complete	STROBE checklist, TRIPOD+AI checklist, pipeline spec, GNN hyperparameters, thresholds, module inventory, feature comparison, full signal cache (6,259 pairs)
6	Data Proof from Live Database	✓ Complete	7 Excel files queried directly from PostgreSQL
7	PLOS ONE Compliance Audit	✓ Complete	7/7 publication criteria met, 16/16 formatting checks passed
8	Figure Generation (Fig 1–9 + S1)	✓ Complete	PNG + TIFF (300 DPI) + EPS formats

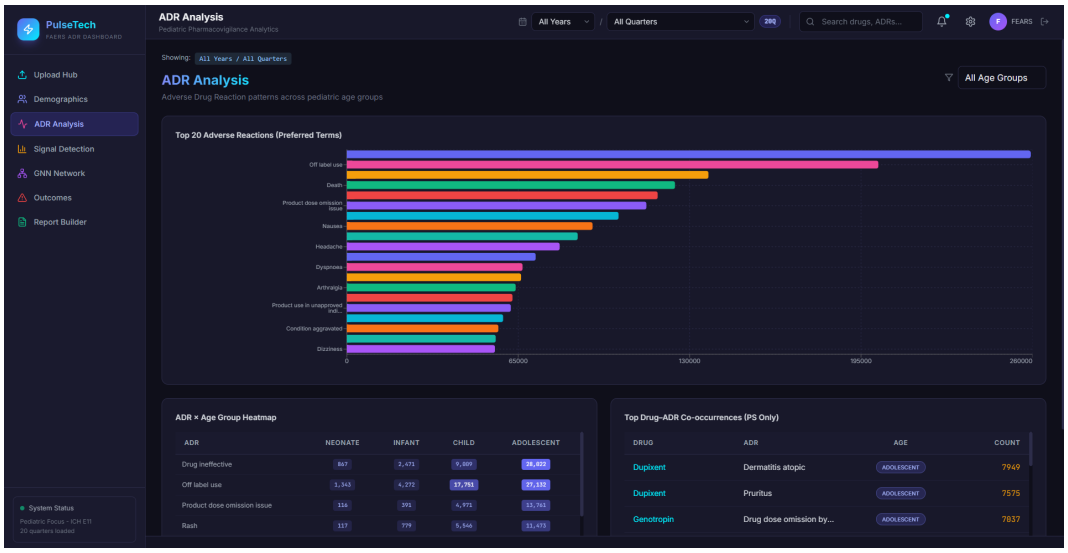


Fig 5 — ADR Analysis: Top-20 adverse reactions, ADR × Age Group heatmap, drug–ADR co-occurrences

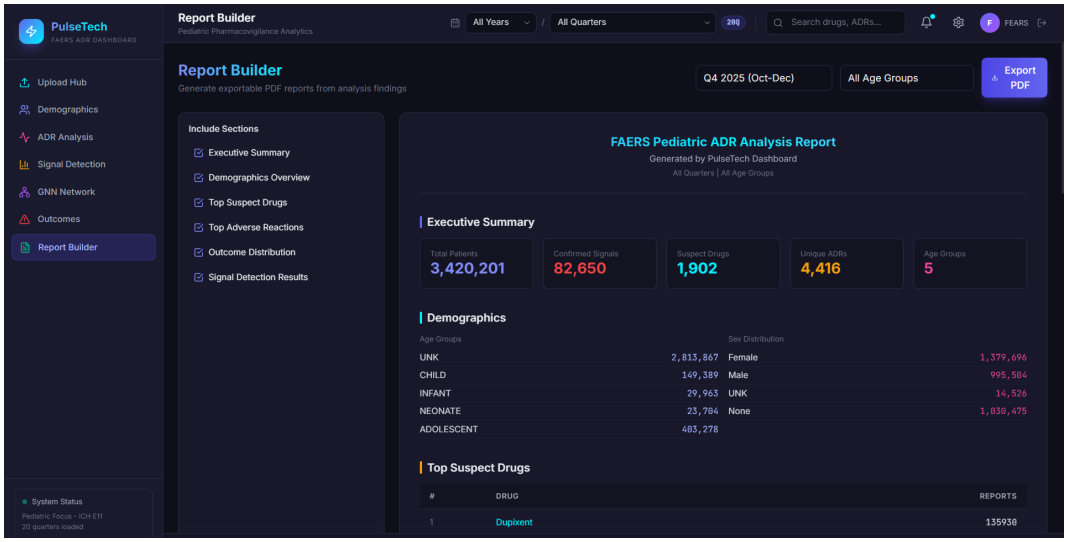


Fig 6 — Report Builder: section-selectable PDF export with live preview

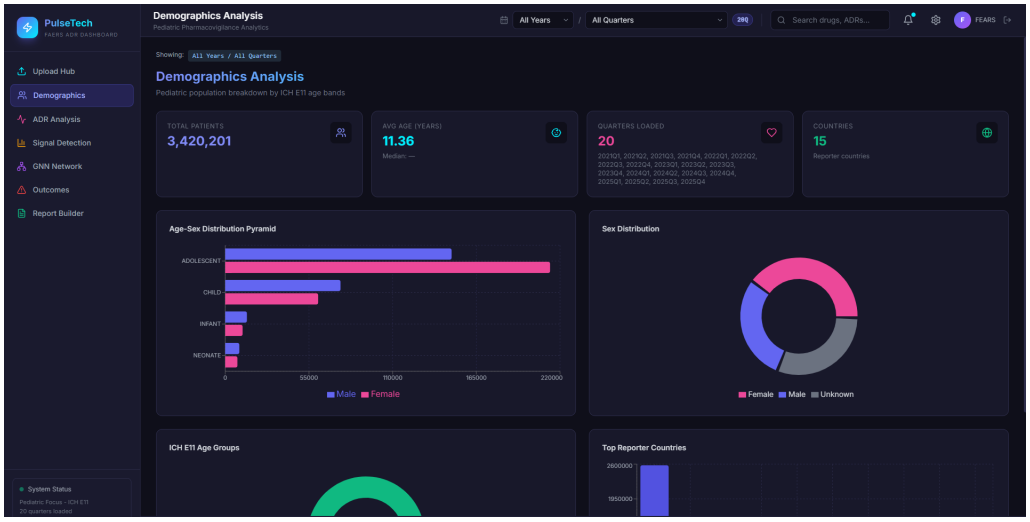
5. Current System Status

Component	Status	%
13-Step FAERS Ingestion Pipeline	✓ Fully Operational	100%
PostgreSQL Database (20 quarters)	✓ Fully Loaded	100%
RxNorm Drug Normalisation	✓ Operational (~94%)	100%
Upload Hub Module	✓ Complete	100%
Demographics Module	✓ Complete	100%
ADR Analysis Module	✓ Complete	100%
Signal Detection (PRR/ROR/IC/EBGM)	✓ Complete	100%
HANConv GNN Model	✓ Trained & Deployed	100%
GNN Network View Module	✓ Complete	100%
Outcomes & Severity Module	✓ Complete	100%
Report Builder (PDF Export)	✓ Complete	100%
PLOS ONE Research Paper	✓ Draft Complete	100%
Supporting Information (S1–S8)	✓ Complete	100%
Formal Usability Study	■ Planned	0%

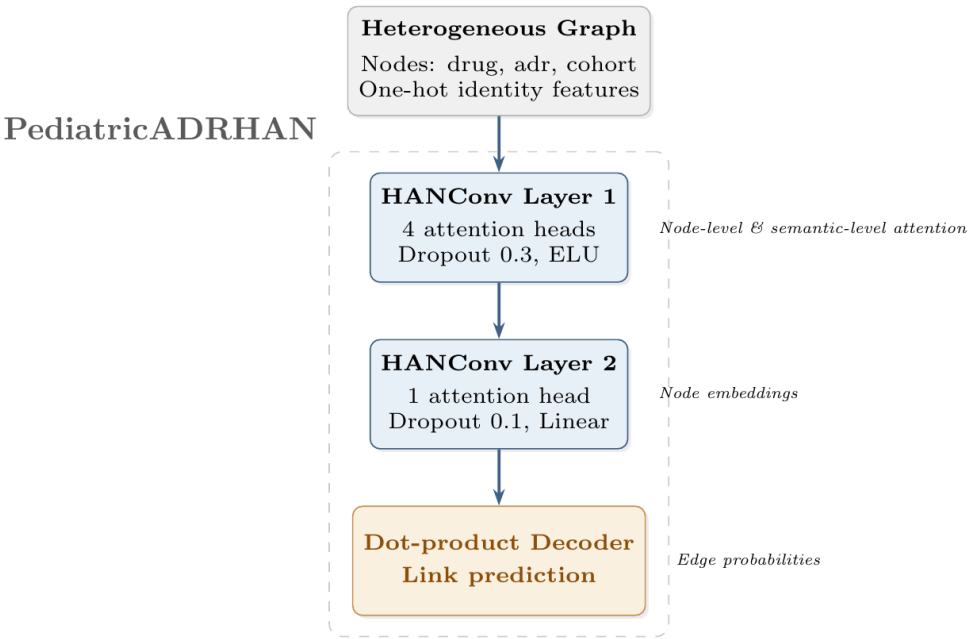
6. Key Metrics & Achievements

Metric	Value
Total Patient Records	3,420,201
FAERS Quarters Loaded	20 (2021 Q1 – 2025 Q4)
Unique Suspect Drugs	1,902
Unique ADR Preferred Terms	4,416
Confirmed Disproportionality Signals	82,650
ICH E11(R1) Pediatric Age Bands	5
GNN Graph — Drug Nodes (CHILD)	79
GNN Graph — ADR Nodes (CHILD)	259
GNN Graph — Edges	500
Signal Edge Density	83.8% (419 / 500)
GNN Architecture	2-layer HANConv, 64-dim, 4-head attention
GNN Training	200 epochs, Adam optimiser
RxNorm Resolution Rate	~94%
Pipeline Runtime	4–12 min per quarter
Dashboard Modules	7 fully operational
Research Paper	PLOS ONE, 845 lines, 32 refs

7. Additional Dashboard Screenshots



Demographics Module — ICH E11(R1) age band distribution with age–sex pyramid



PediatricADRHAN Architecture — 2-layer HANConv with semantic attention layer



Outcomes Module — Seven-category outcome distribution across the 20-quarter corpus

8. Next Steps

#	Planned Task	Target
1	PLOS ONE submission via Editorial Manager	Week 4
2	Cross-age-band GNN benchmarking (Neonate, Infant, Adolescent)	Week 4–5
3	Formal usability evaluation (SUS / heuristic)	Week 5–6
4	VigiBase / EudraVigilance comparison study	Week 6
5	Final project presentation preparation	Week 7

Prepared by: Atharv Telkar & Amey Telkar

Date: 05 / 10 / 2026

Guide Signature: _____